Belief sharing lowers free energy at the project-pool corner

With the standard-Bayes client limit and project-pool recovery identity pinned exactly (sec. 10.1), the study suite opens one step away from them. Its estimand is the paired,

Belief sharing lowers free... (part 2)

seed-level difference in colony-mean variational free energy between incommunicado and communicating conditions.

The design is a categorical source-mechanism analogue of the colony belief-sharing result (Friston et al. 2024).

Belief sharing lowers free... (part 3)

A colony of 7 sentinels observes one hidden location through independent noisy sensors of acuity 0.55. Each agent forms a one-step posterior. The communicating condition applies one standard log-linear-pool round at the `robustness=0` recovery corner (eq. 10, eq. 14);

Belief sharing lowers free... (part 4)

the incommunicado condition retains the paired private posteriors.

Belief sharing lowers free... (part 5)

Under shared support, posterior-log-potential admission, and fixed weights, the pool specializes Eq. 7's message-combination term. It does not reproduce the complete source protocol. The paired comparison gives the following declared quantities:

Belief sharing lowers free... (part 6)

Communicating mean. $\bar{F}_{\text{share}} = 13.2190$ (across-seed 95% bootstrap CI $[12.9656, 13.4685]$ over $n = 480$ seeds). The single illustrative seed-0 run has its own colony mean 15.8115.

Belief sharing lowers free... (part 7)

Its per-agent 95% bootstrap CI is $[15.6177, 16.0647]$ over its $n = 7$ agents. That interval characterizes the displayed run's per-agent spread, not the across-seed mean.

Incommunicado mean. $\bar{F}_{\text{solo}} = 16.5298.$

Belief sharing lowers free... (part 8)

Paired reduction from sharing.
$\Delta \bar{F} = \bar{F}_{\text{solo}} - \bar{F}_{\text{share}} = 3.3109$; positive values mean lower free energy under communication.

Belief sharing lowers free... (part 9)

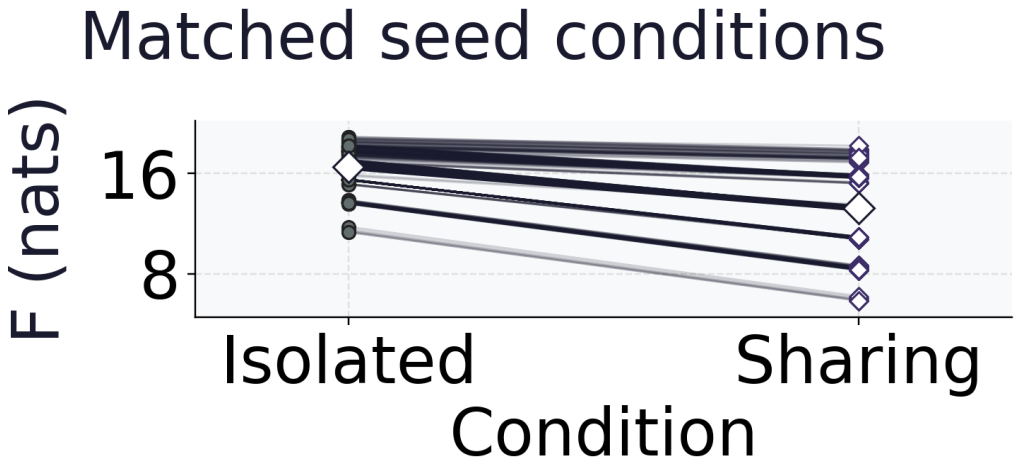
Each seed supplies one matched pair of colony means. Seeds are the independent unit; agents are nested within each seed and are not additional independent replicates. Across the declared seeds, the communicating condition also has higher mean true-state accuracy (0.7302) and lower mean surprise (0.3775).

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Belief sharing lowers free... (part 10)

These are conditional results for the reduced protocol.

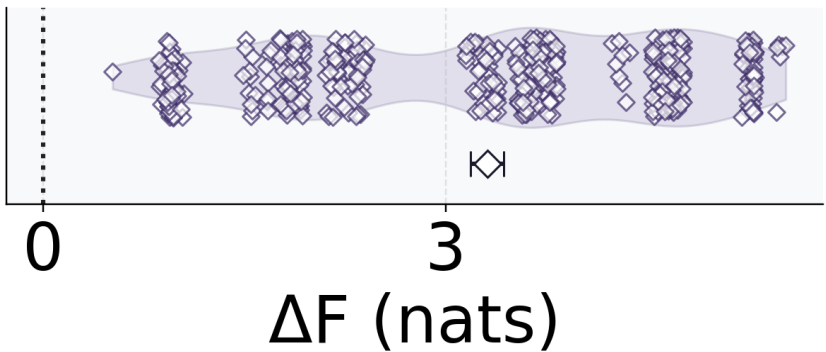
Belief sharing lowers free… (part 11)



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Belief sharing lowers free... (part 12)

Paired seed differences



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Belief sharing lowers free... (part 13)

ΔF = isolated minus communicating.
Positive values indicate lower F
with communication in this reduced
categorical protocol.

Belief sharing lowers free... (part 14)

Filled circles: isolated. Open diamonds: sharing. Fine lines preserve pairing. Dark line / open diamonds: condition means.

Belief sharing lowers free… (part 15)

One open diamond per seed; vertical jitter carries no outcome meaning. Violin is descriptive. Dark dotted rule marks zero contrast.

Belief sharing lowers free... (part 16)

Paired mean ΔF: +3.311 nats.
Independent unit: configured seed; n = 480.

Belief sharing lowers free... (part 17)

Report-owned 95% interval: [3.186, 3.433] nats.

Belief sharing lowers free... (part 18)

Analysis unit: configured seed with within-seed paired conditions. Agents and ordered steps remain nested within seed.

Belief sharing lowers free... (part 19)

Reduced-protocol comparison. It does not establish a general communication benefit or reconstruct the complete source protocol.

Belief sharing lowers free... (part 20)

fig. 12 reports the headline gap; fig. 13 shows the per-agent mechanism behind it.

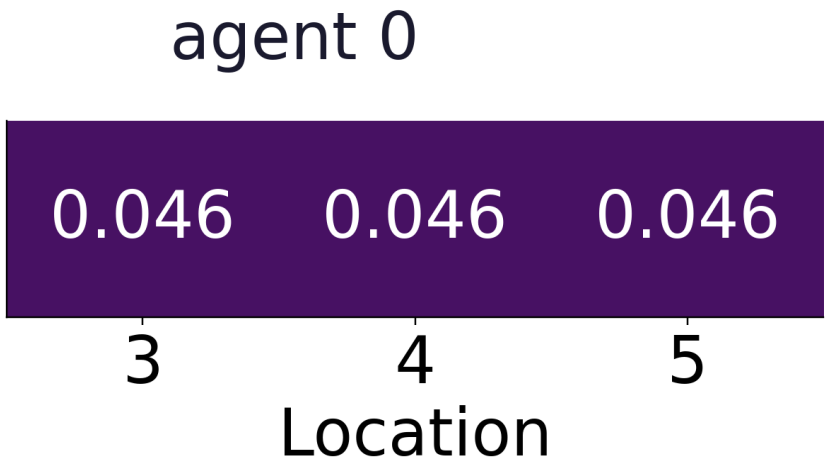
Belief sharing lowers free... (part 21)

Posterior probability mass on a common 0–1 scale. Rows retain agent order, followed by consensus. Location indices retain their original order.

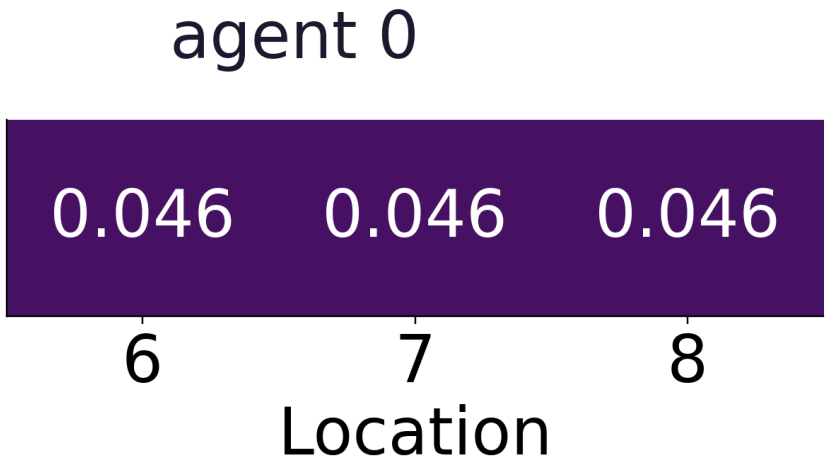
Belief sharing lowers free... (part 22)



Belief sharing lowers free... (part 23)



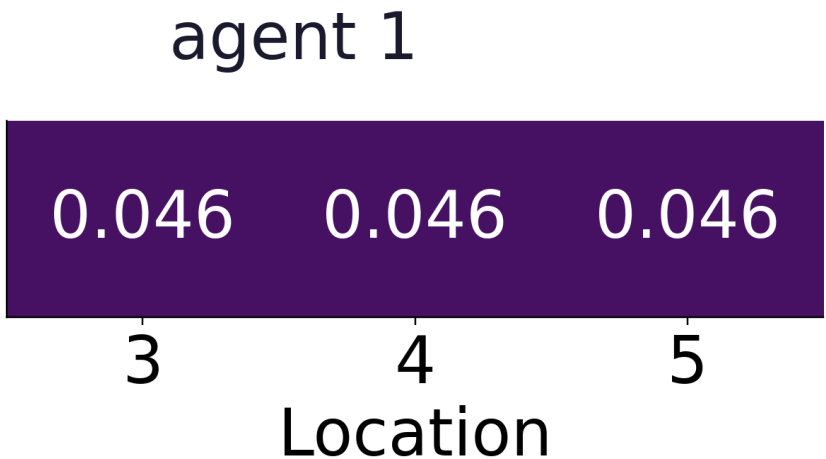
Belief sharing lowers free... (part 24)



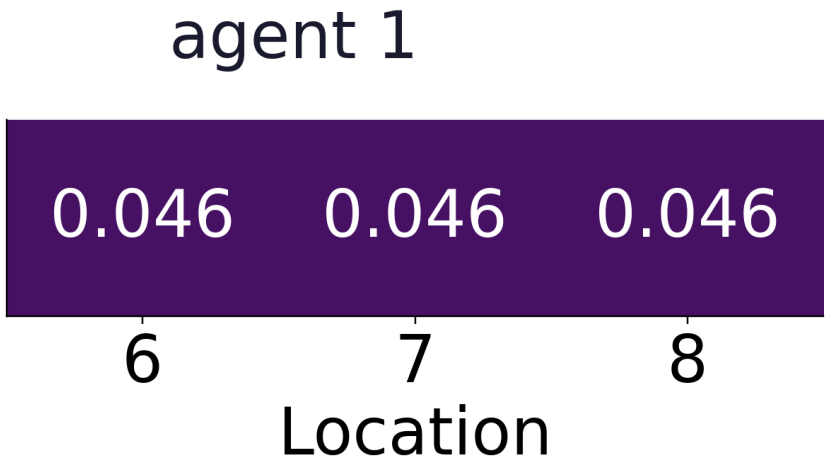
Belief sharing lowers free... (part 25)



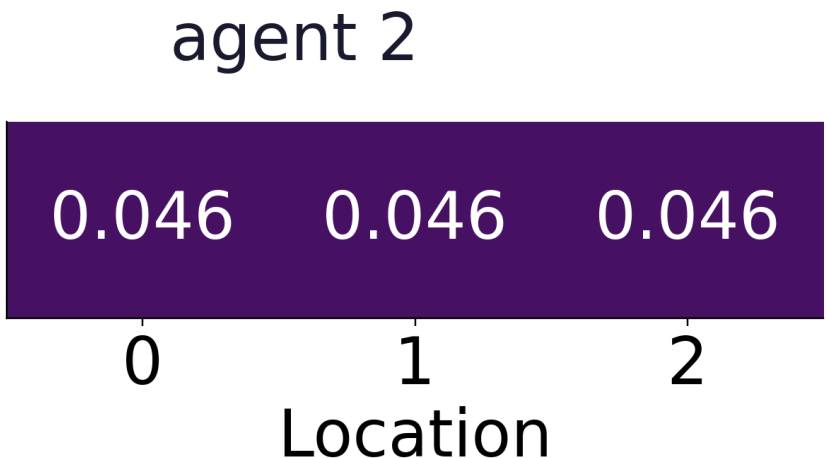
Belief sharing lowers free... (part 26)



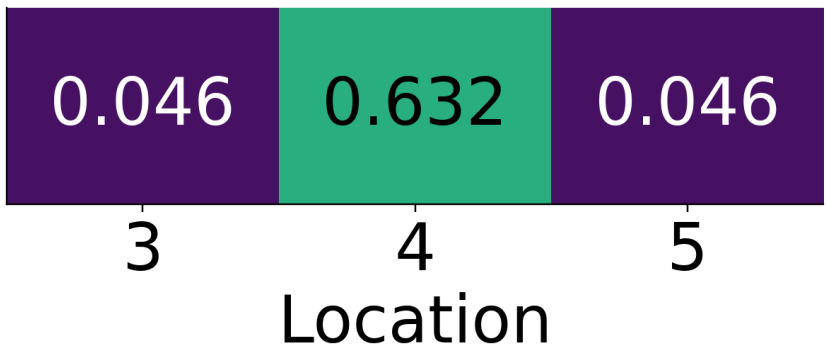
Belief sharing lowers free... (part 27)



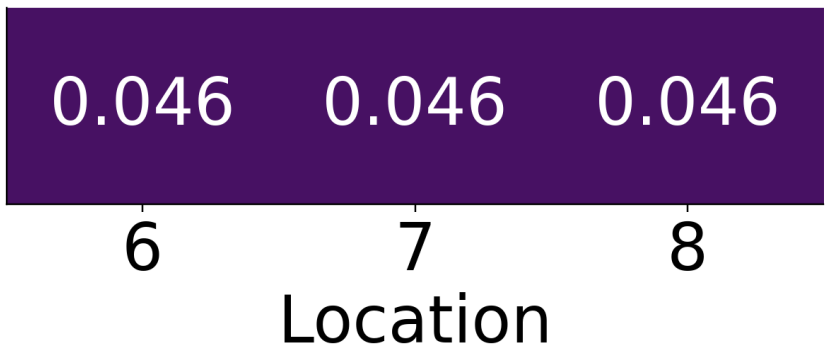
Belief sharing lowers free... (part 28)



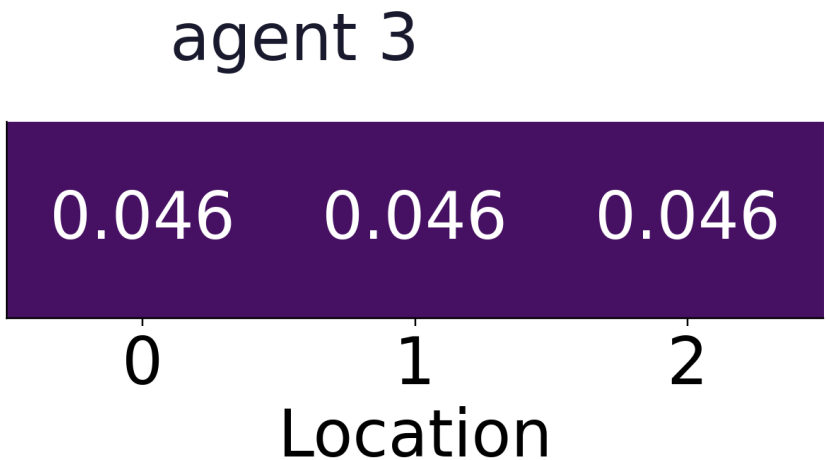
Belief sharing lowers free... (part 29)



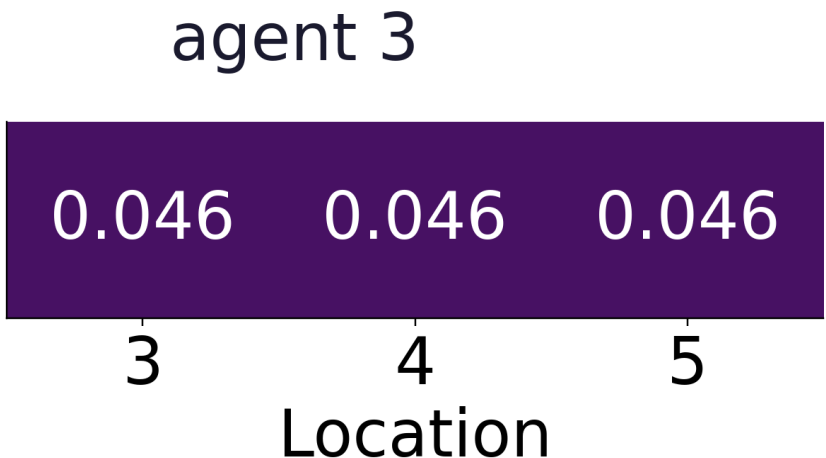
Belief sharing lowers free... (part 30)



Belief sharing lowers free... (part 31)



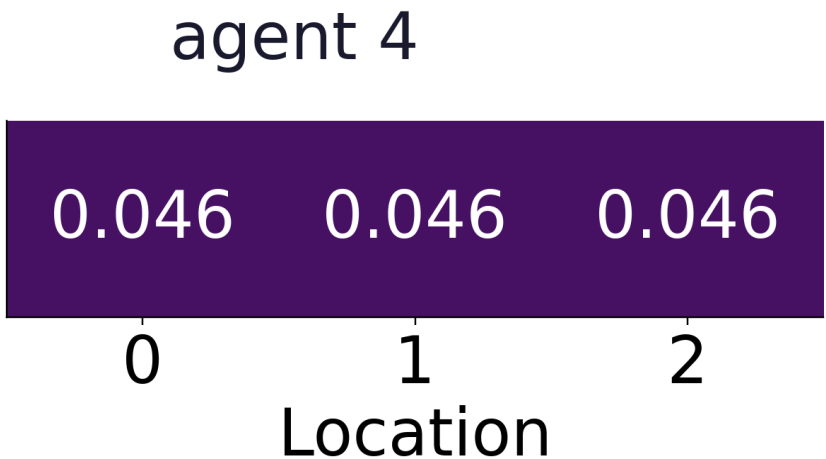
Belief sharing lowers free... (part 32)



Belief sharing lowers free... (part 33)



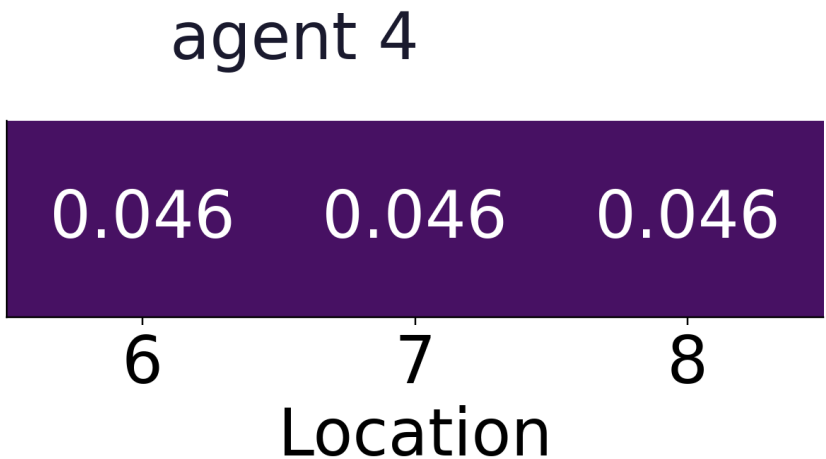
Belief sharing lowers free... (part 34)



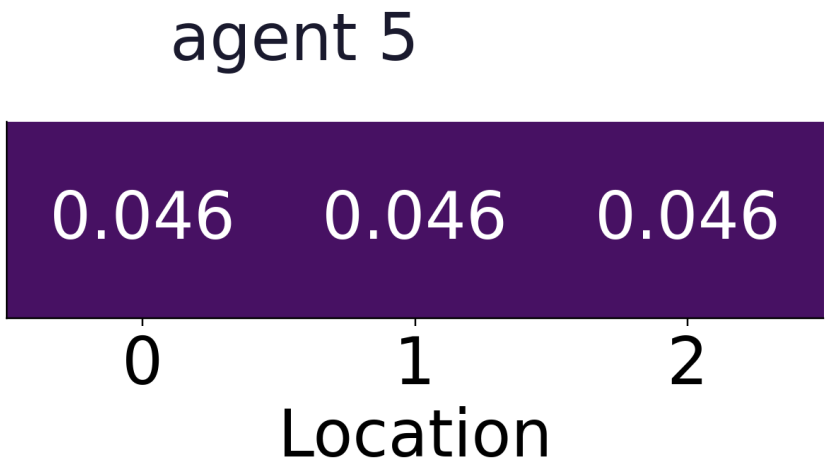
Belief sharing lowers free... (part 35)



Belief sharing lowers free... (part 36)



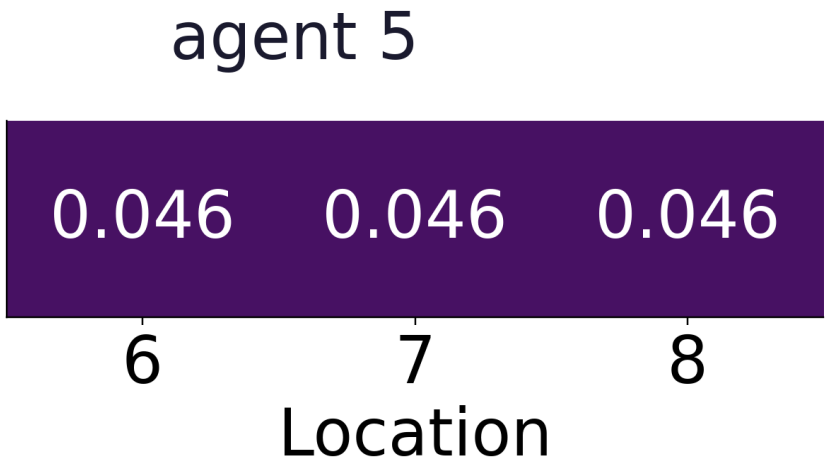
Belief sharing lowers free... (part 37)



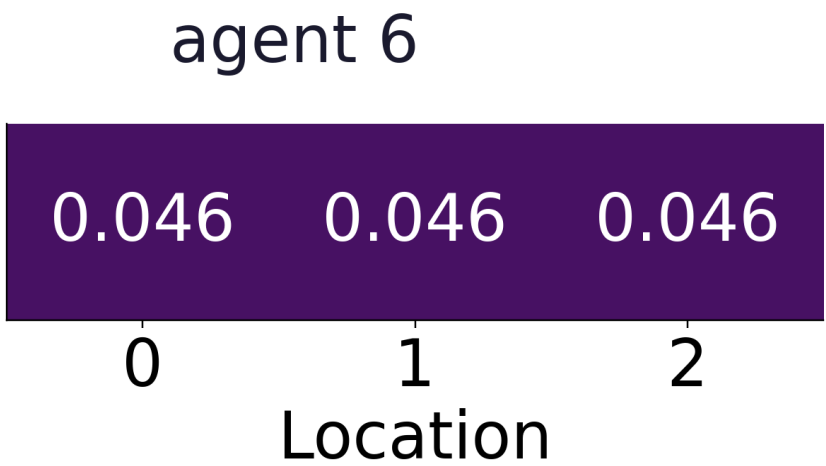
Belief sharing lowers free... (part 38)



Belief sharing lowers free... (part 39)



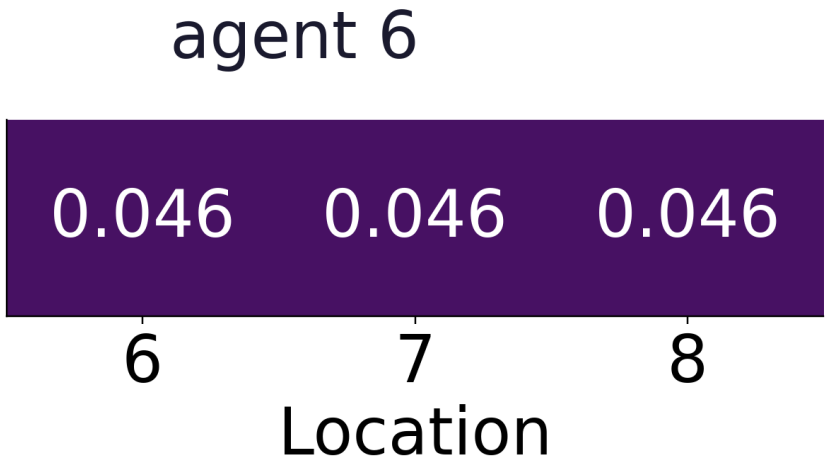
Belief sharing lowers free... (part 40)



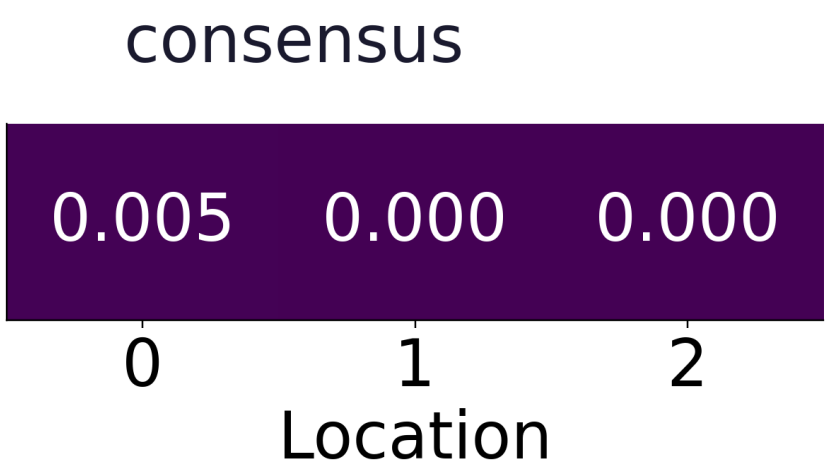
Belief sharing lowers free... (part 41)



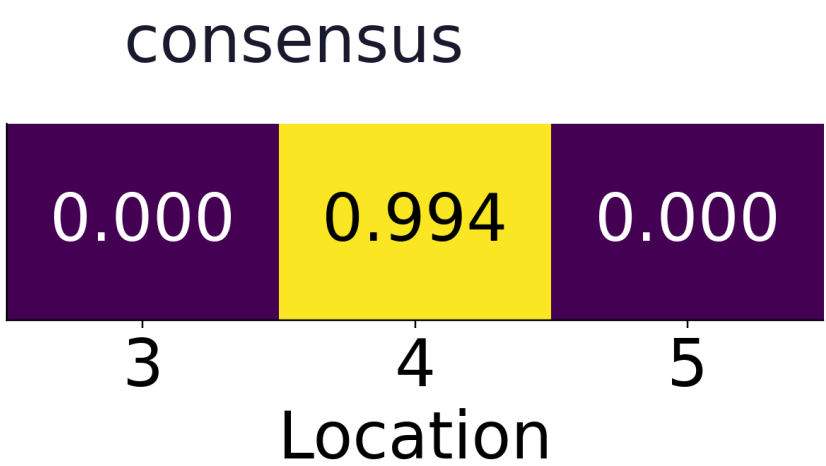
Belief sharing lowers free... (part 42)



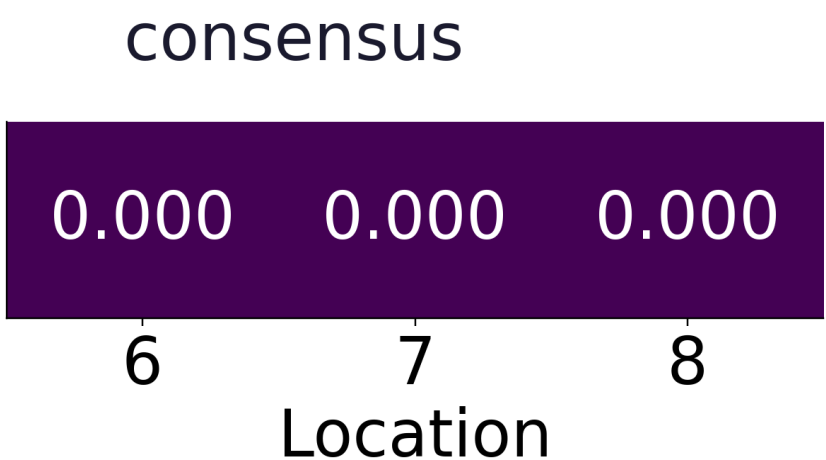
Belief sharing lowers free... (part 43)



Belief sharing lowers free... (part 44)



Belief sharing lowers free... (part 45)



Belief sharing lowers free... (part 46)

7 agents and one consensus row. One configured colony snapshot; no uncertainty interval. Categorical log-linear specialization, not the full source protocol.

Three robustness axes remain distinct in the results

The authoritative three-axis guarantee map is sec. 6.3. The contaminated studies keep its source-conditional client update, heuristic server reweighting,

Three robustness axes... (part 2)

and objective-backed variational server rule visually and inferentially separate. In particular, no result below transfers the client's bounded-influence theorem or the variational rule's raw-weight bound to `robust_aggregate`.